

5G RAN

Coordinated Interference Management (Low-Frequency TDD) Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 02 (2025-08-30).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationship between L/NR Inter-band Deep Coordination and coordinated interference management. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added support for optimized cooperating cell selection. For details, see: • 4.1 Principles • 4.3.2.1 Prerequisite Functions • 4.4.2.1 Data Preparation • 4.4.2.2 Using MML Commands	Modified parameter: Added the COOP_CELL_ALLOC_OPT_SW option to the NRDUCellComp.CompOptSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Added the impact relationships of downlink 8-layer transmission for a single FWA user with the two functions: "interference rejection precoding" and "interference rejection precoding in mobile scenarios". For details, see [4.3.2.3 Function Impacts](#).

Modified the figure that illustrates CS. For details, see [Figure 4-1](#).

Revised the conditions for the two functions to take effect: "interference rejection precoding" and "interference rejection precoding in mobile scenarios". For details, see [4.1 Principles](#).

Revised other descriptions in the document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-051210	Coordinated Interference Management	4 Coordinated Interference Management

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Coordinated interference management	None	4 Coordinated Interference Management

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Coordinated interference management	Supported only in low frequency bands	4 Coordinated Interference Management

3 Overview

Intra-frequency networking is generally used for NR networks to improve spectrum resource utilization. However, user experience of cell edge users (CEUs) is poor in intra-frequency networking due to high inter-cell interference. Coordinated interference management can be used to improve user experience of such UEs. Coordinated interference management allows gNodeBs to fully utilize inter-cell time-frequency or space-domain resources by leveraging coordinated scheduling (CS) and coordinated beamforming (CBF), reducing inter-cell interference and improving user experience of CEUs.

Table 3-1 describes the concepts related to coordinated interference management.

Table 3-1 Related concepts

Name	Definition
Serving cell	Cell where a UE accesses
Cooperating cell	Intra-frequency neighboring cell that works with the serving cell for joint processing
Overlapping area	Coverage overlap between cells
CS/CBF UE	UE that suffers interference from neighboring cells in the overlapping area, also referred to as victim UE or beneficiary UE
Interfering UE	UE that is served by a cooperating cell and causes interference to CS/CBF UEs

4 Coordinated Interference Management

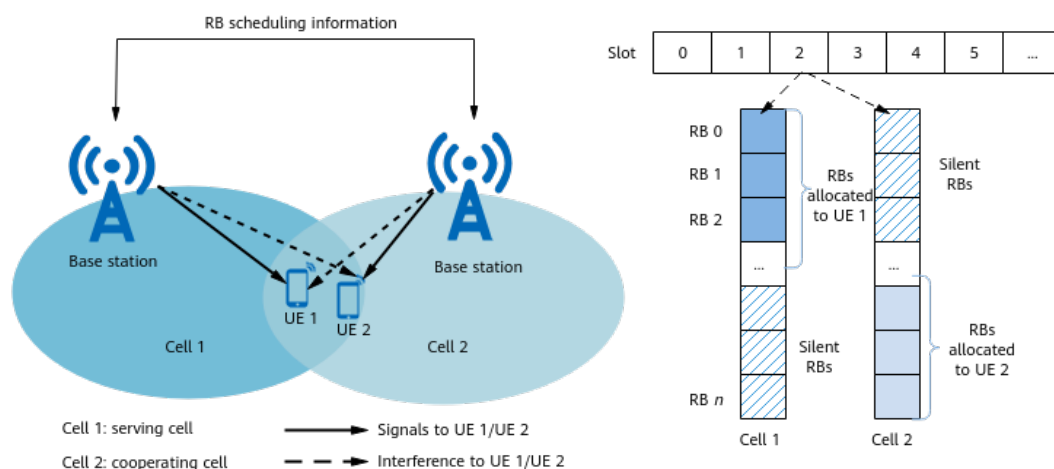
4.1 Principles

CS/CBF can work between intra-frequency cells, either within the same base station or across base stations. The **INTRA_GNB_DL_CS_CBF_SW** and **INTER_GNB_DL_CS_CBF_SW** options of the **NRDUCellAlgoSwitch.CompSwitch** parameter control intra-base-station CS/CBF and inter-base-station CS/CBF, respectively. Further details are as follows:

- CS, an interference coordination function that works through frequency-domain resource coordination

This function mitigates interference for UEs in the overlapping area (referred to as victim UEs) by coordinating frequency-domain resource positions for UE scheduling between the serving and cooperating cells, thereby improving the performance of these UEs. For this function to work, there must be one and only one UE engaged in a large-packet service in each of the serving and cooperating cells. Examples of large-packet services include video, gaming, and file downloads.

Figure 4-1 CS

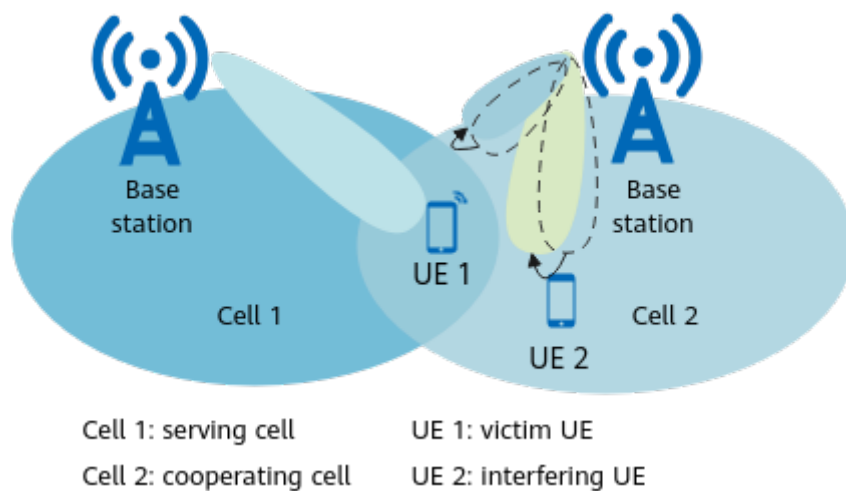


- CBF, a beam-domain-based interference coordination function

A cooperating cell adjusts the weight for an interfering UE, redirecting the interfering beam to significantly mitigate interference for a victim UE while keeping the loss within a controllable range for the interfering UE. Consequently, the performance of demodulation improves for the victim UE, ultimately improving user experience. CBF requires that victim UEs engage in large-packet services.

Fast CBF, which is controlled by the **QUICK_CBF_SW** option of the **NRDUCellComp.CsCbfAlgoSwitch** parameter, benefits more UEs compared to CBF. When fast CBF is enabled, UEs other than those engaged in large-packet services can also benefit from CBF. Fast CBF can be enabled only when the **USER_CHARACTER_SRS_ADAPT_SW** option of the **NRDUCellSrs.SrsAlgoSwitch** parameter is selected.

Figure 4-2 CBF



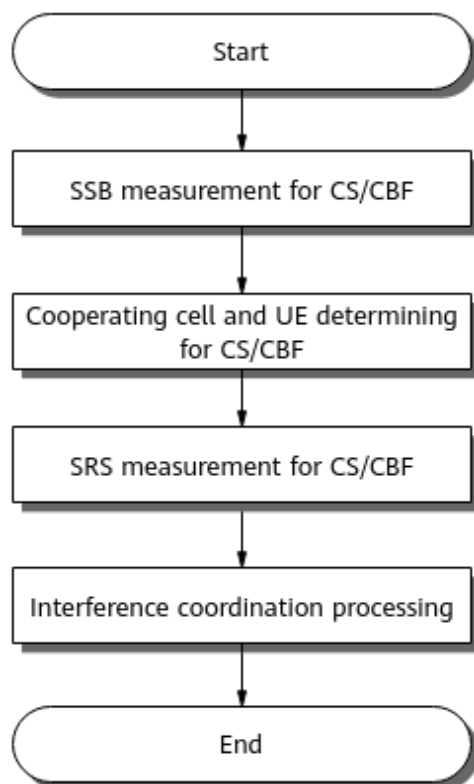
Currently, coordinated interference management applies to the following scenarios:

- Between common cells
- Between a combined cell and a common cell
- Between a combined cell in joint transmission mode and a common cell
- Between a distributed massive MIMO cell and a common cell

Coordinated Interference Management Between Common Cells

Figure 4-3 illustrates the processing that occurs when an interfering UE and a victim UE are both served by common cells (with the **NRDUCell.NrDuCellNetworkingMode** parameter set to **NORMAL_CELL**).

Figure 4-3 Interference coordination between common cells



1. SSB measurement for CS/CBF

The UE starts SSB RSRP measurement after setting up a service and sends an event A3 measurement report when a certain condition is met.

Event A3 indicates that the signal quality of a neighboring cell becomes offset better than that of the serving cell. A UE reports event A3 if the entering condition $(Mn + Ofn + Ocn - Hys > Ms + Ofs + Ocs + Off)$ is met throughout *TimeToTrigger*. The variables are described as follows:

- *TimeToTrigger* indicates the period in which the condition is met. *TimeToTrigger* for event A3 in CS/CBF is specified by the **NRCellComp.TimeToTrigger** parameter.
- *Ms* and *Mn* are the measurement results of the serving and neighboring cells, respectively.
- *Ofs* and *Ofn* are the frequency offsets of the serving and neighboring cells, respectively. The frequency offsets between intra-frequency cells are 0.
- *Ocs* and *Ocn* are the cell individual offsets (CIOs) of the serving and neighboring cells, respectively. *Ocs* is set to 0. *Ocn* is specified by the **NRCellRelation.CellIndividualOffset** parameter, which is configured during neighboring cell planning. When this function is enabled, neighboring cell replanning is not required, and therefore, the parameter value does not need to be adjusted.
- *Hys* is the hysteresis for the measurement result. *Hys* for event A3 in CS/CBF is specified by the **NRCellComp.Hysteresis** parameter.

- *Off* is the offset for the measurement result. *Off* in the formula for event A3 in CS/CBF is specified by the **NRCellComp.DlCompRsrpOffset** parameter.

For details about event A3, see section 5.5 "Measurements" in 3GPP TS 38.331 V15.5.0.

2. Cooperating cell and UE determining for CS/CBF

The gNodeB verifies whether the received event A3 measurement report contains neighboring cells that meet both of the following conditions:

- CS/CBF is enabled.
- All of the following parameters have the same values as the serving cell:

- **NRDUCell.NrDuCellNetworkingMode**
- **NRDUCell.FrequencyBand**
- **NRDUCell.DlNarfcn**
- **NRDUCell.DlBandwidth**
- **NRDUCell.SlotAssignment**

If the report contains such neighboring cells, up to three of them with the best signal quality can be selected as cooperating cells for CS/CBF, and the UE that sends the event A3 measurement report is identified as a CS/CBF UE. If the report does not contain such neighboring cells, no further action is required.

In dense cell deployments, there may be no neighboring cell suitable for selection as a cooperating cell. If this occurs, CS/CBF cannot work. To increase the probability of CS/CBF working, it is recommended that the optimized cooperating cell selection function be enabled (by selecting the **COOP_CELL_ALLOC_OPT_SW** option of the **NRDUCellComp.CompOptSwitch** parameter). This function allows cooperating cells to be selected from a wider range of cells for UEs in the overlapping area.

3. SRS measurement for CS/CBF

Separate SRS measurements are initiated in the serving and cooperating cells of the identified CS/CBF UE. If none of the selected cooperating cells can offer the SRS resources corresponding to those allocated in the serving cell for the UE, CBF cannot take effect.

4. Interference coordination processing

Further CS/CBF processing relies on the CS/CBF UE's SRS measurements in the serving and cooperating cells. If the absolute difference in SRS RSRP between the serving and cooperating cells is less than the **NRDUCellComp.CsCbfSrsRsrpOffset** parameter value and the SINR of SRS in the cooperating cell is greater than the **NRDUCellCollabServ.CbfSrsPreSinrThld** parameter value, further CS/CBF processing takes place, as described below:

- If the channel correlation between the CS/CBF UE and an interfering UE in the cooperating cell is greater than the **NRDUCellComp.CsCbfChannelCorrThld** parameter value, whether CS can be executed is determined. CS can be executed only when both of the following conditions are met:

- The MCS index of the CS/CBF UE is less than the **NRDUCellComp.CsMcsThld** parameter value.
- The estimated ratio of the overall spectral efficiency after CS to that before CS is greater than the **NRDUCellComp.CsSpctEfficiencyThld** parameter value.
- If the channel correlation between the CS/CBF UE and an interfering UE in the cooperating cell is less than or equal to the **NRDUCellComp.CsCbfChannelCorrThld** parameter value, whether CBF can be executed is determined. CBF can be executed only when specific conditions are met, as described below:
 - Conditions that must all be met for CBF to be executed if the SINR of the interfering UE in the cooperating cell is greater than 20 dB:
 - The CS/CBF UE's cooperating-cell channel frequency selective factor is less than the **NRDUCellComp.CbfUeFreSelFactorThld** parameter value.
 - The estimated spectral efficiency loss for the interfering UE in CBF mode is less than 25%.
 - The estimated overall spectral efficiency improvement through CBF is greater than the **NRDUCellComp.CbfSpecEfficiencyJudgeThld** parameter value. For fast CBF to be executed, the following additional condition must be met: The estimated ratio of the overall spectral efficiency in fast CBF mode to the spectral efficiency in single-cell mode is greater than the **NRDUCellComp.QuickCbfSpctEffThld** parameter value.
 - Conditions that must be met for CBF to be executed if the SINR of the interfering UE in the cooperating cell is less than or equal to 20 dB:

The estimated overall spectral efficiency improvement through CBF is greater than the **NRDUCellComp.CbfSpecEfficiencyJudgeThld** parameter value. For fast CBF to be executed, the following additional condition must be met: The estimated ratio of the overall spectral efficiency in fast CBF mode to the spectral efficiency in single-cell mode is greater than the **NRDUCellComp.QuickCbfSpctEffThld** parameter value.

Coordinated Interference Management Between a Combined Cell and a Common Cell

If a UE is served by a combined cell (with **NRDUCell.NrDuCellNetworkingMode** set to **HYPER_CELL_COMBINE_MODE** and **NRDUCellMultiTrp.DataTransMode** set to **BASIC_MODE** or **DPS_MODE**), a common cell (with **NRDUCell.NrDuCellNetworkingMode** set to **NORMAL_CELL**) working in 32T32R mode or higher can be selected as a cooperating cell for coordinated interference management, with the same processing as between common cells. Fast CBF is not supported in this scenario. The TRPs in the combined cell must work in the same TX/RX mode, which must be 2T2R, 4T4R, or 8T8R.

Coordinated Interference Management Between a Combined Cell in Joint Transmission Mode and a Common Cell

If a UE is served by a combined cell in joint transmission mode (with **NRDUCell.NrDuCellNetworkingMode** set to **HYPER_CELL_COMBINE_MODE** and **NRDUCellMultiTrp.DataTransMode** set to **JOINT_MODE**), a common cell (with **NRDUCell.NrDuCellNetworkingMode** set to **NORMAL_CELL**) working in 32T32R mode or higher can be selected as a cooperating cell for coordinated interference management, with the same processing as between common cells.

Coordinated Interference Management Between a Distributed Massive MIMO Cell and a Common Cell

If a UE is served by a distributed massive MIMO cell (with the **DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch.DmMimoSwitch** parameter selected), a common cell (with **NRDUCell.NrDuCellNetworkingMode** set to **NORMAL_CELL**) working in 32T32R mode or higher can be selected as a cooperating cell for coordinated interference management, with the same processing as between common cells.

NOTE

For details about combined cells in joint transmission mode and combined cells, see *Cell Combination*. For details about distributed massive MIMO cells, see *Distributed Antenna Solutions (Low-Frequency TDD)*.

Enhanced Functions

If a cooperating cell is a low-frequency TDD cell in 32T32R or higher mode, the following functions can be enabled to further increase the downlink throughput of UEs that benefit from CBF:

- Interference rejection precoding: When the SINR of SRS from an interfering UE is greater than or equal to the **NRDUCellPdsch.SrsPresinrThld** parameter value, this function optimizes the weight for that UE to further mitigate interference for beneficiary UEs, thereby improving the downlink throughput of these beneficiary UEs. This function is controlled by the **DL_MU_NLOS_IR_PRECODING_SW** option of the **NRDUCellAlgoSwitch.MuMimoSwitch** parameter and can be enabled only when the **USER_CHARACTER_SRS_ADAPT_SW** option of the **NRDUCellSrs.SrsAlgoSwitch** parameter is selected. LampSite base stations do not support this function.
- Interference rejection precoding in mobile scenarios: When the SINR of SRS from a mobile interfering UE is greater than or equal to the **NRDUCellPdsch.SrsPresinrThld** parameter value, this function optimizes the weight for that UE to further mitigate interference for mobile beneficiary UEs, thereby improving the downlink throughput of these beneficiary UEs. This function is controlled by the **DL_MU_MOBILE_IR_PRECODING_SW** option of the **NRDUCellAlgoSwitch.MuMimoSwitch** parameter. LampSite base stations do not support this function.

4.2 Network Analysis

4.2.1 Benefits

In intra-frequency continuous networking, CS/CBF can be used to reduce interference to traffic channels of UEs in the overlapping area and improve downlink throughput.

For details about how to evaluate the gains brought by this function, see [4.4.4 Network Monitoring](#).

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

CBF slightly reduces the downlink throughput of interfering UEs served by cooperating cells, as it prevents these UEs from interfering with beneficiary UEs served by the serving cell. Despite this reduction, the overall gains delivered by this function remain positive.

When coordinated interference management takes effect during a handover, the following changes may be observed in drive tests: increased handover delay, decreased SSB RSRP, and fluctuation in the values of indicators such as the block error rate and throughput in both the uplink and downlink.

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051210	Coordinated Interference Management	NR0S00CDIM00	Per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
SRS interference coordination based on self-contained slots	SRS_INTRF_COORD_S_SLOTT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	None	Before you enable optimized cooperating cell selection (by selecting the COOP_CELL_ALLOC_OPT_SW option of the NRDUCellComp.CompOptSwitch parameter), SRS interference coordination based on self-contained slots must be enabled.

4.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Coordinated interference management is not supported in Hyper Cell networking.
U-TDOA-based positioning/TDOA fingerprint-based positioning	UTDOA_SW option of the NRCCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	None
SRS field strength-based positioning	SRS_RSRP_LOCATION_SW option of the NRCCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	None
Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	The two functions cannot be both enabled: Fusion Cell and coordinated interference management.

Function Name	Function Switch	Reference	Description
LTE TDD and NR Spectrum Sharing Switch	LTE_NR_TDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	None	None
CCE partial symbol adaptation	CCE_PARTIAL_SYM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	The two functions cannot be both enabled: fast CBF and CCE partial symbol adaptation.

4.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Power saving BWP	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	After the COOR_UE_PWR_SAVE_PRI_ADJ_SW option of the NRDUCellCollabServ.CommAlgoSwitch parameter is selected, fast CBF UEs can have faster downlink throughput. However, they may spend a smaller proportion of time in power saving mode (specifically, in BWP2 mode), which is indicated by the decrease in the value of the N.Bwp2.Dur.Total counter. This lessens the power saving effect for the UEs.

Function Name	Function Switch	Reference	Description
Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	For UEs involved in intra-base-station DL CoMP, CS and CBF are not performed.
Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	For UEs involved in Co-MM, CS and CBF are not performed.
External interference avoidance	NRDUCellIntrfIdent.IntrfIdentificationMethod set to UL_MANUAL or UL_AUTO	<i>Interference Avoidance</i>	After external interference avoidance is enabled, SRS is not transmitted within the range where interference is present. As a result, the probability that CS and CBF take effect decreases.
Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	RB0 may not be the start RB for scheduling UEs within the minimum RB quota in a cooperating cell when both of the following conditions are met: (1) Coordinated interference management is enabled; (2) The start RB of the minimum RB quota (for resource reservation) is configured. (Currently, only RB0 can be configured as the start RB.)

Function Name	Function Switch	Reference	Description
NR DU cell resource management between network slices	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.NetworkSliceAlgoSwitch parameter	<i>Network Slicing</i>	RB0 may not be the start RB for scheduling network slice UEs within the minimum RB quota in a cooperating cell when both of the following conditions are met: (1) The two functions—'NR DU cell resource management between network slices' and 'coordinated interference management'—are both enabled; (2) The start RB of the minimum RB quota is configured for the cooperating cell. (Currently, only RB0 can be configured as the start RB.)
SRS remote interference avoidance	SRS_RIM_INTRF_AVOID_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When SRS remote interference avoidance is enabled in the serving cell but disabled in cooperating cells, coordinated interference management does not take effect.

Function Name	Function Switch	Reference	Description
PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	CBF is not performed in slots with MU-MIMO pairing. If the proportion of MU-MIMO pairing is high, the probability that CBF takes effect decreases.
High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	For high-speed cells, CBF does not take effect.
DRX	BASIC_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter	<i>DRX</i>	- After DRX is enabled, the probability that CS and CBF take effect decreases. - After DRX is enabled, it is recommended that the FULL_BUFFER_UE_EXIT_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter be selected for both the serving and cooperating cells.
LTE TDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	When 'coordinated interference management' and 'LTE TDD and NR Flash Dynamic Power Sharing' are both enabled, the power sharing gains may decrease on the NR side.

Function Name	Function Switch	Reference	Description
L/NR Inter-band Deep Coordination	LNR_INTER_BAND_DEEP_COORD_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	When L/NR Inter-band Deep Coordination and coordinated interference management are both enabled, the power sharing gains may decrease on the NR side.
NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When coordinated interference management and NR inter-carrier dynamic power sharing are both enabled, the benefits of the latter may decrease.
Cell-level MRO	MRO_SCENARIO_IDENT_SW option of the gNB Mro.ScenarioIdentifySwitch parameter and INTRA_FREQ_MRO_SW option of the NRCellMro.MroSwitch parameter	<i>MRO</i>	If coordinated interference management and cell-level intra-frequency MRO are both enabled, event A3 reporting in coordinated interference management is affected. As a result, the identification of UEs in overlapping coverage areas in coordinated interference management is affected.

Function Name	Function Switch	Reference	Description
UE-level MRO	UE_MRO_SWITCH option of the NRCellMro.MroSwitch parameter	<i>MRO</i>	If coordinated interference management and UE-level MRO are both enabled and coordinated interference management takes effect for ping-pong UEs, event A3 reporting in coordinated interference management is affected. As a result, the identification of UEs in overlapping coverage areas in coordinated interference management is affected.
Downlink type 1 scheduling for slice groups with the TTI-based control policy	DL_TYPE1_SCH_SW option of the NRDUCellRes.SchedulingPolicySwitch parameter	<i>Network Slicing</i>	When downlink type 1 scheduling for slice groups with the TTI-based control policy takes effect, coordinated interference management does not take effect for UEs in the slice groups.
PDCCH skipping	PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter	<i>UE Power Saving</i>	When both coordinated interference management and PDCCH skipping are enabled, PDCCH skipping does not take effect on CS/CBF UEs.

Function Name	Function Switch	Reference	Description
Downlink MU anti-interference scheduling	DL_MU_ANTI_INTRF_SCH_SW option of the NRDUCellDLMimo.DLMuMimoSchSupplementSw parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Downlink MU anti-interference scheduling does not take effect in slots where CBF or fast CBF takes effect.
LTE and NR multi-band power adaptive scheduling	LNR_MULTI_BAND_PWR_ADPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	When 'LTE and NR multi-band power adaptive scheduling' and 'coordinated interference management' are both enabled, the gains of power adaptive scheduling may decrease on the NR side.
NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When coordinated interference management and NR multi-band power adaptive scheduling are both enabled, the gains provided by the latter function may decrease.
Intelligent power control energy saving	SMART_PWR_CTRL_ENERGY_SAVE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Intelligent power control energy saving does not take effect for CS/CBF UEs.
Rank optimization for cell-edge UEs	FAR_UE_RANK_OPT_SW option of the NRDUCellDLMimo.DLMuMimoSchSupplementSw parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Rank optimization for cell-edge UEs does not apply to UEs for which CBF or fast CBF is in effect.

Function Name	Function Switch	Reference	Description
Inter-cell interference avoidance	INTER_CELL_INTRF_AVOID_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Fast CBF does not apply to UEs for which inter-cell interference avoidance is in effect.
Downlink precise RB estimation	PRECISE_DL_RB_ESTIMATE_SW option of the NRDUCellDISch.DLSchAlgorithmSwitch parameter	<i>Downlink Scheduling</i>	When enabled with downlink precise RB estimation, coordinated interference management may produce reduced benefits.
Positioning-dedicated SRS transmission in uplink slots	SRS_FOR_POSITION_ON_USLOT_SW option of the NRDUCellPosAlgo.LcsMeasureAlgoSw parameter	<i>Positioning</i>	When both positioning-dedicated SRS transmission in uplink slots and coordinated interference management are enabled: If a UE does not support positioning-dedicated SRS transmission, coordinated interference management cannot take effect for the UE. If a UE supports positioning-dedicated SRS transmission, coordinated interference management can take effect for the UE.

Function Name	Function Switch	Reference	Description
Downlink 8-layer transmission for a single FWA user	DL_SU_EIGHT_LAYER_SW option of the NRDUCellDLRank.DIRankSelAlgoSw parameter	<i>FWA</i>	If downlink 8-layer transmission for a single FWA user has taken effect, "interference rejection precoding in mobile scenarios" and "interference rejection precoding" do not take effect for the user.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- For a macro cell, all NR-capable main control boards and NR TDD-capable baseband processing units support this function.
- For details about the board requirements for a combined cell in joint transmission mode and a combined cell, see *Cell Combination*.
- For details about the board requirements for a distributed massive MIMO cell, see *Distributed Antenna Solutions (Low-Frequency TDD)*.

RF Modules

- For a macro cell, all NR TDD-capable RF modules that comply with the eCPRI protocol and work in 32T32R mode or higher support this function; all NR TDD-capable RF modules that adopt the 8T8R mode or lower support this function.
- For details about the RF module requirements for a combined cell in joint transmission mode and a combined cell, see *Cell Combination*.
- For details about the RF module requirements for a distributed massive MIMO cell, see *Distributed Antenna Solutions (Low-Frequency TDD)*.

4.3.4 Cells

- The **NRDUCell.DuplexMode** parameter must be set to **CELL_TDD**.
- CBF requires that cooperating cells be common cells (**NRDUCell.NrDuCellNetworkingMode** set to **NORMAL_CELL**) that work in 32T32R mode or higher.
- Distributed massive MIMO cells cannot function as cooperating cells in CBF.

4.3.5 Others

CS has no requirements for UEs. CBF has no requirements for interfering UEs, but requires that beneficiary UEs be antenna selection UEs.

In CS/CBF, the beneficiary UEs and interfering UEs must belong to the same operator.

Inter-base-station CS/CBF requires that gNodeBs be connected using the eXn interface and the end-to-end delay be less than or equal to 4 ms. Fast CBF requires that the end-to-end delay be less than or equal to 0.5 ms. For details about how to configure the eXn interface and how to check whether the eXn interface is successfully set up, see *eXn Self-Management*.

4.4 Operation and Maintenance

4.4.1 Precautions

Throughput-related counters for UEs in the overlapping area are measured only when the **INTRA_GNB_OVERLAPPING_MEAS_SW** or **INTER_GNB_OVERLAPPING_MEAS_SW** option of the **NRDUCellAlgoSwitch.CompSwitch** parameter is selected. In addition, the **NRCellComp.OverlappingRsrpOffset** parameter needs to be configured, and its value must be equal to that of either **NRCellComp.UlCompRsrpOffset** or **NRCellComp.DlCompRsrpOffset**.

To ensure precise estimation of the gains of CS/CBF, the **NRCellComp.OverlappingRsrpOffset** and **NRCellComp.DlCompRsrpOffset** parameters must be set to the same value.

For details about counters, see [4.4.4 Network Monitoring](#). To compare the throughput before and after CS/CBF is enabled, it is good practice to perform the preceding configurations one week before CS/CBF is enabled. This ensures that sufficient statistics can be collected for comparison purposes. In common cell networking, to evaluate whether to enable CS/CBF, monitor the average number of UEs in RRC_CONNECTED mode in the overlapping area through the **N.User.RRCCConn.Avg.CellOverlap** counter.

4.4.2 Data Configuration

4.4.2.1 Data Preparation

Table 4-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
CoMP Switch	NRDUCellAlgoSwitch.CompSwitch	- INTRA_GNB_DL_CS_CBF_SW - INTER_GNB_DL_CS_CBF_SW	To enable CS/CBF, select both options in the serving and neighboring cells to be involved in interference coordination.
CoMP Switch	NRDUCellAlgoSwitch.CompSwitch	- INTRA_GNB_OVERLAPPING_MEAS_SW - INTER_GNB_OVERLAPPING_MEAS_SW	It is recommended that both options be selected one week before CS/CBF is enabled to observe the throughput changes before and after CS/CBF is enabled.
DL CoMP RSRP Offset	NRCCellComp.DlCompRsrpOffset	None	Retain the default value.
Overlapping RSRP Offset	NRCCellComp.OverlappingRsrpOffset	None	It is recommended that this parameter be set to the value of NRCCellComp.DlCompRsrpOffset .
Hysteresis	NRCCellComp.Hysteresis	None	Retain the default value.
Time to Trigger	NRCCellComp.TimeToTrigger	None	Retain the default value.
CS/CBF SRS RSRP Offset	NRDUCellComp.CsCbfSrsRsrpOffset	None	Retain the default value. For sites that have been upgraded from a version earlier than V100R005C10, manually set this parameter to 30 (default value).
CBF SRS PreSINR Threshold	NRDUCellCollabServ.CbfSrsPreSinrThld	None	Retain the default value.
CS/CBF Channel Correlation Threshold	NRDUCellComp.CsCbfChannelCorrThld	None	Retain the default value.

Parameter Name	Parameter ID	Option	Setting Notes
CS MCS Threshold	NRDUCellComp.CsMcsThld	None	Retain the default value. For sites that have been upgraded from a version earlier than V100R005C10, manually set this parameter to 23 (default value).
CS Spectral Efficiency Threshold	NRDUCellComp.CsSpctEfficiencyThld	None	Retain the default value.
CBF UE Freq Sel Factor Threshold	NRDUCellComp.CbfUeFreqSelFactorThld	None	Retain the default value.
CBF Spectral Efficiency Judge Threshold	NRDUCellComp.CbfSpecEfficiencyJudgeThld	None	Retain the default value.
DL Link Adaptation Algorithm Switch	NRDUCellPdsch.DLLinkAdaptAlgoSwitch	DL_TD_MCS_ADJ_SW	Select this option so that more accurate MCS indexes can be selected for UEs with CS taking effect, increasing the downlink rate of these UEs.
CS/CBF Algorithm Switch	NRDUCellComp.CsCbfAlgoSwitch	QUICK_CBF_SW	Use the recommended value.
Quick CBF Spectral Efficiency Threshold	NRDUCellComp.QuickCbfSpctEffThld	None	Use the recommended value.
MU-MIMO Switch	NRDUCellAlgoSwitch.MuMimoSwitch	DL_MU_NLOS_IR_PRECODING_SW	It is recommended that this option be selected to increase the downlink throughput of UEs that benefit from CBF.
MU-MIMO Switch	NRDUCellAlgoSwitch.MuMimoSwitch	DL_MU_MOBILE_IR_PRECODING_SW	It is recommended that this option be selected to increase the downlink throughput of UEs that benefit from CBF.
CoMP Optimization Switch	NRDUCellComp.CompOptSwitch	COOP_CELL_ALLOC_OPT_SW	It is recommended that this option be selected to increase the probability of intra-base-station DL CoMP taking effect.

4.4.2.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

You are advised to turn on specific switches and configure related parameters (as described below) one week before CS/CBF is enabled to collect UE throughput in the overlapping area. This facilitates a comparison of throughput before and after CS/CBF is enabled.

```
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
CompSwitch=INTRA_GNB_OVERLAPPING_MEAS_SW-1&INTER_GNB_OVERLAPPING_MEAS_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
CompSwitch=INTRA_GNB_OVERLAPPING_MEAS_SW-1&INTER_GNB_OVERLAPPING_MEAS_SW-1;
MOD NRCELLCOMP: NrCellId=0, OverlappingRsrpOffset=-18, DlCompRsrpOffset=-18,
TimeToTrigger=320MS, Hysteresis=2;
MOD NRCELLCOMP: NrCellId=1, OverlappingRsrpOffset=-18, DlCompRsrpOffset=-18,
TimeToTrigger=320MS, Hysteresis=2;
```

Activation Command Examples

```
//Enabling CS/CBF
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
CompSwitch=INTRA_GNB_DL_CS_CBF_SW-1&INTER_GNB_DL_CS_CBF_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
CompSwitch=INTRA_GNB_DL_CS_CBF_SW-1&INTER_GNB_DL_CS_CBF_SW-1;
//Setting CS/CBF-related thresholds
MOD NRDUCELLCOMP: NrDuCellId=0, CsCbfSrsRsrpOffset=30, CsCbfChannelCorrThld=16, CsMcsThld=23,
CsSptEfficiencyThld=110, CbfUeFreSelFactorThld=8, CbfSpecEfficiencyJudgeThld=10;
MOD NRDUCELLCOMP: NrDuCellId=1, CsCbfSrsRsrpOffset=30, CsCbfChannelCorrThld=16, CsMcsThld=23,
CsSptEfficiencyThld=110, CbfUeFreSelFactorThld=8, CbfSpecEfficiencyJudgeThld=10;
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, CbfSrsPreSinrThld=-100;
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, CbfSrsPreSinrThld=-100;
//Enabling downlink time-domain MCS adjustment
MOD NRDUCELLPDSCH: NrDuCellId=0, DLinkAdaptAlgoSwitch=DL_TD_MCS_ADJ_SW-1;
MOD NRDUCELLPDSCH: NrDuCellId=1, DLinkAdaptAlgoSwitch=DL_TD_MCS_ADJ_SW-1;
//Setting the QuickCbfSptEffThld parameter
MOD NRDUCELLCOMP: NrDuCellId=0, QuickCbfSptEffThld=70;
MOD NRDUCELLCOMP: NrDuCellId=1, QuickCbfSptEffThld=70;
//Enabling interference rejection precoding
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, MuMimoSwitch=DL_MU_NLOS_IR_PRECODING_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, MuMimoSwitch=DL_MU_NLOS_IR_PRECODING_SW-1;
//Enabling interference rejection precoding in mobile scenarios
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, MuMimoSwitch=DL_MU_MOBILE_IR_PRECODING_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, MuMimoSwitch=DL_MU_MOBILE_IR_PRECODING_SW-1;
//Enabling optimized cooperating cell selection
MOD NRDUCELLCOMP: NrDuCellId=0, CompOptSwitch=COOP_CELL_ALLOC_OPT_SW-1;
MOD NRDUCELLCOMP: NrDuCellId=1, CompOptSwitch=COOP_CELL_ALLOC_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling CS/CBF
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
```

```
CompSwitch=INTRA_GNB_DL_CS_CBF_SW-0&INTER_GNB_DL_CS_CBF_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
CompSwitch=INTRA_GNB_DL_CS_CBF_SW-0&INTER_GNB_DL_CS_CBF_SW-0;
//Turning off the switches for intra-base-station overlapping area measurement to reduce signaling
exchange over the air interface (recommended when the throughput of UEs in the overlapping area does
not need to be observed)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
CompSwitch=INTRA_GNB_OVERLAPPING_MEAS_SW-0&INTER_GNB_OVERLAPPING_MEAS_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
CompSwitch=INTRA_GNB_OVERLAPPING_MEAS_SW-0&INTER_GNB_OVERLAPPING_MEAS_SW-0;
```

4.4.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.3 Activation Verification

On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management > NR > Cell Performance Monitoring > CS/CBF Statistic(Cell) Monitoring**. Check the values of **CS Schedule Users Num** and **CBF ZeroForcing RB Num**. If the value of **CS Schedule Users Num** is not 0 for both the serving and cooperating cells, CS has taken effect. If the value of **CBF ZeroForcing RB Num** is not 0 for the cooperating cell, CBF has taken effect.

4.4.4 Network Monitoring

To evaluate the gains of CS/CBF, you can compare the values of the following performance indicators before and after this function is enabled. Given that the indicator values depend on parameter settings, set related parameters before the evaluation by referring to [4.4.1 Precautions](#).

- Downlink throughput of UEs in the overlapping area, which will increase

$$\text{Downlink throughput of UEs in the overlapping area (Mbit/s)} = \frac{(\text{N.ThpVol.DL.CellOverlap} - \text{N.ThpVol.DL.LastSlot.CellOverlap})}{\text{N.ThpTime.DL.RmvLastSlot.CellOverlap}} \times 1000$$
Table 4-2 lists the related performance counters.
- User Downlink Average Throughput (DU)**, which will not decrease

Table 4-2 Downlink performance counters for UEs in the overlapping area

Counter ID	Counter Name
1911820651	N.ThpVol.DL.CellOverlap
1911823024	N.ThpVol.DL.LastSlot.CellOverlap
1911823026	N.ThpTime.DL.RmvLastSlot.CellOverlap

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- *CoMP*
- *Co-MM (Low-Frequency TDD)*
- *UE Power Saving*
- *URLLC*
- *Hyper Cell*
- *Cell Combination*
- *High Speed Mobility*
- *Remote Interference Management (Low-Frequency TDD)*
- *Interference Avoidance*
- *Network Slicing*
- *MIMO (TDD)*
- *DRX*
- *LTE and NR Power Allocation*
- *NR Inter-Carrier Dynamic Power Allocation*
- *eXn Self-Management*
- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *Fusion Cell (Low-Frequency TDD)*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- *MRO*
- *Positioning*
- *License Control Item Lists*
- *Energy Conservation and Emission Reduction*
- *Downlink Scheduling*
- *FWA*